Satisfaction depends only on the interpretations of the symbols occurring in the formula

The Open Logic Project contributors

openlogic:first-order-logic/syntax-and-semantics:local-snapshot-20261013; CC BY 4.0

Editorial note. The counted statement is the author's own proposition environment prop:extensionality, which carries the optional title Extensionality (source lines 883-892); the proof environment begins two lines below it and is its own proof, closing at source line 899.

Editorial note. No \olref, \Olref, \cref or \cite occurs anywhere inside the retained span, so the package declares neither a source pointer nor a bibliography entry.

Editorial note. The author's proof is written as an induction specification (first on terms, then on formulas); the retained package reproduces it exactly as written and does not expand it.

Editorial note. The retained preamble is the minimal working preamble shipped with the frozen OLP unit; note that the frozen preamble's shorthand handling is still defective upstream (the author's !A notation is typeset literally), which affects the rendering of this card in the same way as every other OLP card and is scheduled for a batch re-assembly once the intake line ships the fix.

Editorial note. Section and theorem counters are reset to zero so the retained proposition prints with the numbering of the retained front matter rather than continuing the frozen file's count.

Editorial scope. One original result of the Open Logic Project's first-order syntax and semantics unit, counted once: satisfaction is extensional in the interpretations of the symbols occurring in the formula.

Allowed standard prerequisites (editorial).

- Structure and variable assignment. A first-order structure fixes a domain together with interpretations of the constant, relation and function symbols; a variable assignment maps variables into the domain.. The two structures and the shared assignment are the setting of the statement (source lines 883-888).
- Value of a term and satisfaction of a formula. The value of a term in a structure under an assignment, and the satisfaction relation between a structure, a formula and an assignment, both defined by recursion on the syntax.. The two inductions follow exactly these recursions (source lines 894-899).

Proposition 0.1 (Extensionality)**.** *Let* $!A$ *be !!aformula, and* $\mathfrak{A}M_1$ *and* $\mathfrak{A}M_2$ *be !!structures with* $|\mathfrak{A}|M_1 = |\mathfrak{A}|M_2$*, and* s *a variable assignment on* $|\mathfrak{A}|M_1 = |\mathfrak{A}|M_2$*. If* $scM_1 = scM_2$*,* $sRM_1 = sRM_2$*, and* $sfM_1 = sfM_2$ *for every !!constant* c*, relation symbol* R*, and !!function* f *occurring in* $!A$*, then* $M_1 \models !A[s]$ *iff* $M_2 \models !A[s]$*.*

Proof. First prove (by induction on t) that for every term, $\mathrm{val}(t)M_1[s] = \mathrm{val}(t)M_2[s]$. Then prove the proposition by induction on $!A$, making use of the claim just proved for the induction basis (where $!A$ is atomic). □